SOLOW MODEL

Introduction

The Solow-Swan model (also known as the Solow growth model or exogenous growth model) is a foundational framework in macroeconomics for understanding long-run economic growth. This model explains how economies grow over time through capital accumulation, population/labour growth, and technological progress. It addresses key questions like:

- Why do some countries grow faster than others?
- What determines steady-state income levels?
- Can sustained growth occur without continuous technological improvement?

The Solow model introduces flexible capital-output ratios and exogenous technological change. It predicts that economies converge to a steady state where per capita output grows only at the rate of technological progress, not through endless capital accumulation.

Before the Solow model, economic thought often struggled to explain long-run stability. Solow introduced a "neoclassical" approach, moving away from fixed proportions of labour and capital. He proposed that an economy's output is a result of combining physical capital (K) and labour (L), augmented by the effectiveness of technology (A).

The basic Production Function is expressed as:

$$Y_t = F(K_t, L_t) = A{L_t}^{1-\alpha}{K_t}^{\alpha} \qquad (0<\alpha<1)$$

Where

$Y_t = Output\ at\ time\ t$
$K_t = Capital\ stock\ at\ time\ t$
$L_t = Labot\ at\ time\ t$
$A = Available\ technology\ (measured\ by\ total\ factor\ productivity)$
$\alpha = Capital's\ share\ of\ output (typically\ 0.3 - 0.4)$

The Solow model explains the following key mechanisms:

- **Capital accumulation:-** New investment increases the capital stock, but **depreciation** (wear and tear) constantly reduces it.
- **Diminishing Returns:** As an economy adds more capital to a fixed amount of labour, the "extra" output produced by each new unit of capital decreases. This is why you cannot grow an economy infinitely just by building more factories.
- **The Steady State:** This is the "equilibrium" point where new investment exactly offsets depreciation and population growth. Once an economy reaches this state, growth in income per worker stops unless there is technological progress.
- **Convergence:**- Poorer countries (lower initial K) grow faster and converge to the steady state (conditional convergence, depending on saving, population growth, etc.)
- **Growth Accounting:**- Long-run growth is driven primarily by technological progress (exogenous in the basic model), not just capital accumulation.
- **Policy Insights:**- Increasing saving or reducing population growth raises steady-state income, but only technology drives perpetual per-capita.

Existing Literature

1. **Foundation:-** It introduced a neoclassical production function, where output depends on capital, labour, and exogenous technology. The model shows growth comes from capital accumulation and technological progress, economies converge to a steady state, long-run growth is driven by technology (exogenous)
2. **Augment Solow Model (Human Capital):-** The Human capital significantly improves the model's explanatory power. Differences in income across countries can be explained by **Saving rate, Population growth, Human capital.**
   The empirical studies confirm that Higher saving ->Higher income per capita and Higher population growth ->lower income per capita.
3. **Cross-Country Empirical Literature:-** A large body of literature applies the Solow model to compare countries. The differences in income across countries arise due to **Capital accumulation, Technology differences** and Technology plays a **dominant role** in explaining income differences.
   Convergence Hypothesis:- Poor countries grow faster than rich ones (conditional convergence). However, absolute convergence is mostly observed among similar economies.
4. **Empirical Evidence on Growth Determinants:-** Empirical studies using Solow-type regressions show investment rate has a **positive effect on GDP per capita**, population growth has a **negative effect**, human capital improves long-run growth, technology explains residual differences across countries. Some newer studies also suggest that moderate population growth may sometimes support growth (non-linear effect)

Data and Methodology

1. **Data Source:-** The data used in this study has been collected from the World Bank Open Data, which provides reliable and internationally comparable macroeconomic indicators. The dataset covers four countries—India, China, Japan, and South Korea—over a selected time period. The following data.
   The following variable have been used:
   - Gross Domestic Product (GDP) – to measure total economic output
   - Labour Force – Proxy for total workers in the economy
   - Gross Domestic Saving (GDS) – User to approximate saving behavior
   - Gross Capital Formation (GCF) – used as a proxy for investment
2. **Construction of Variables:-** To apply the Solow Growth Model, the raw data has been transformed into per-worker terms and capital dynamics.
   a) **Output per Worker**

$$y = \frac{GDP}{Labour\ Force}$$

   b) **Investment per Worker**

$$i = \frac{GCF}{Labour\ Force}$$

   c) **Labour Growth Rate(n)**

$$n = \frac{L_t - L_{t-1}}{L_{t-1}}$$

   d) **Capital per Worker**

Initial Capital per Worker

$$k_0 = \frac{i_0}{(\delta + n)}$$

This assumes the economy starts at a steady – state approximation.

For subsequent periods, Capital per worker evolves as:

$$k_{t+1} = (1 - \delta - n)k_t + i_t$$

This Equation shows that capital increases through investment and capital decreases due to depreciation and labour growth.

e) Depreciation of Capital

Depreciation is assumed to be constant δ = 0.05 (5% per year). The effective depreciation per worker is given by:

$$(\delta + n)K$$

This accounts for the both physical depreciation and dilution of capital due to labour growth.

f) Model Framework

The study is based on the Solow Growth Model, where output per worker is determines by:

$$y_t = Ak_t^{\alpha}$$

Where $y_t = Output\ per\ Worker$

$A = Available\ technology\ (measured\ by\ total\ factor\ productivity)$

$k_t = Capital\ per\ worker$

$\alpha = Capital's\ share\ of\ output (typically\ 0.3 - 0.4)$

g) Steady State Capital per worker

The steady state, the point at which capital per worker k_t comes to rest and stops changing, occurs when $\Delta k_t = 0$

$$k^* = \frac{i}{(\delta + n)}$$

Findings and Conclusions

On the basis of our graphical analysis we find the why China, Japan, South Korea grew fast than the India.

Japan and South Korea:- In my graph show actual capital per worker (k) rising steadily for decades. These countries maintained extremely high investment rates that moved from a very low starting point toward a high – income steady state. These countries show very high levels of capital per worker. South Korea shows highest capital accumulation, indicating strong investment led growth.

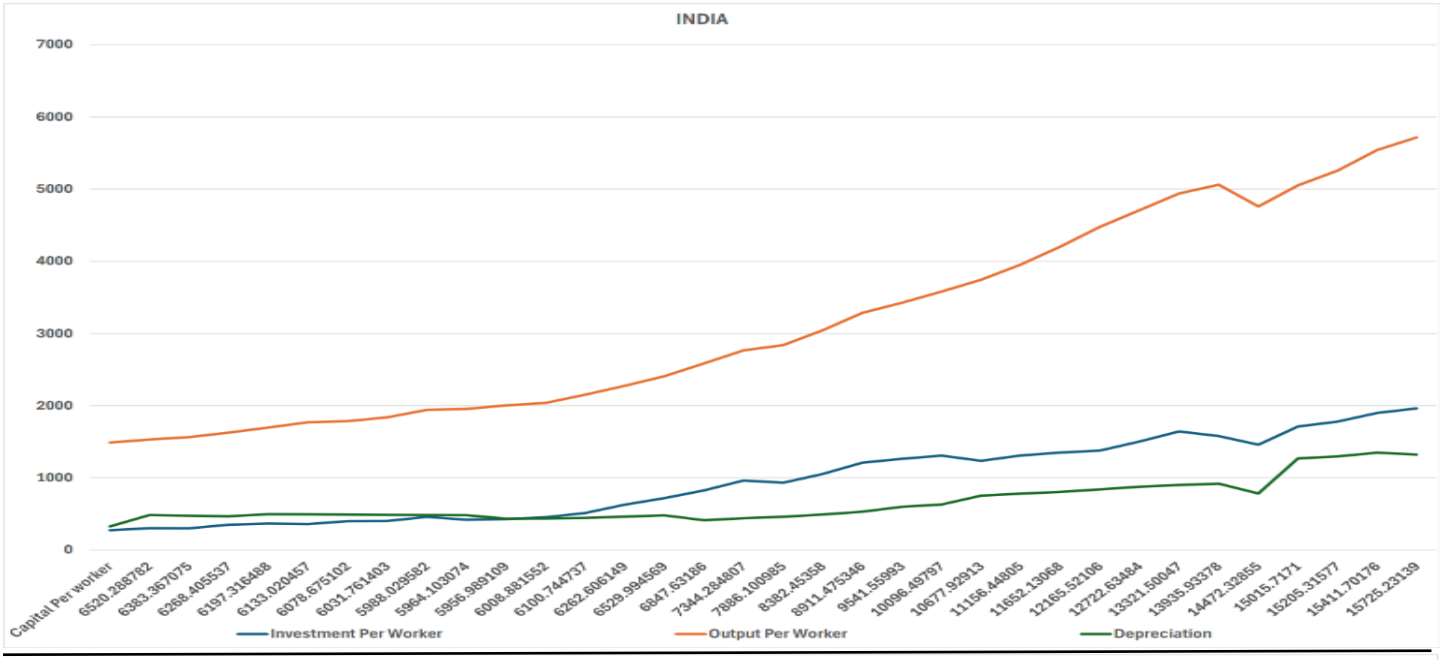
China:- In the Chin's graph show a dramatic explosion in both "Output Per Worker" and "Investment" starting in the early 2000s. Unlike India, China's investment per worker grew at an almost exponential rate, allowing it to bypass the "diminishing returns" phase for much longer than the model predicts.

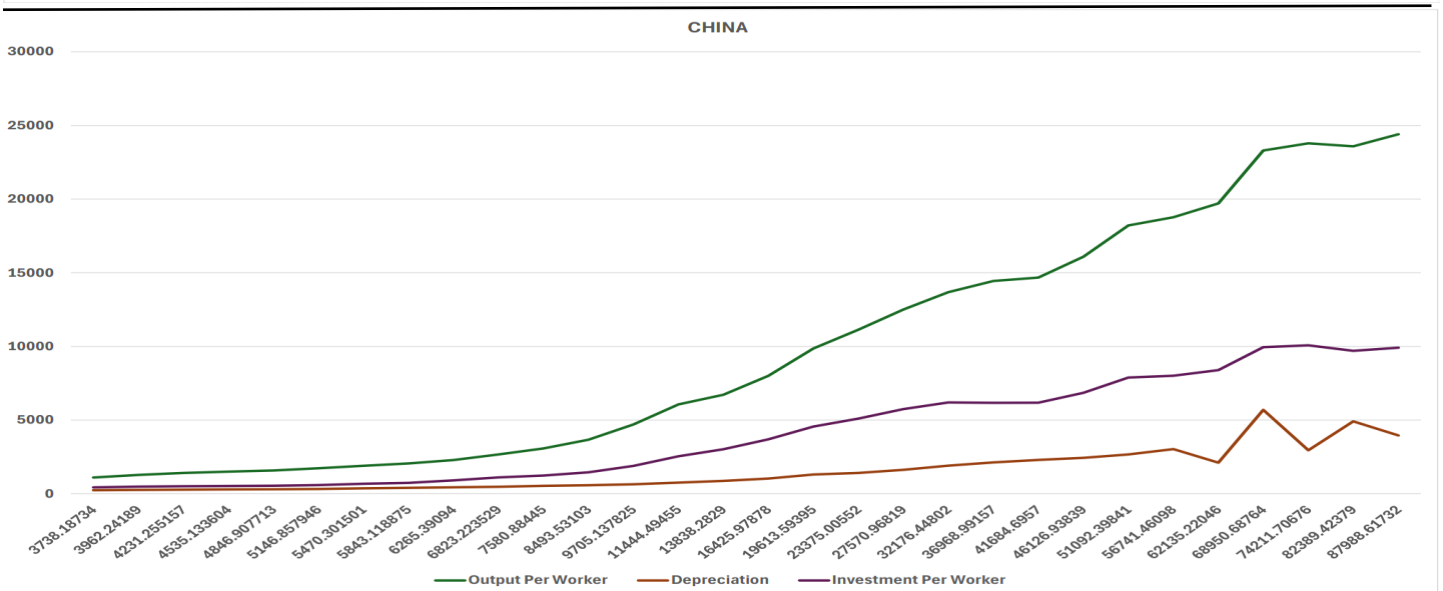
India:- India's lower average investment rate (s)—historically constrained by pre-1991 policies, lower domestic savings, and higher population growth rate (n)—produces a lower steady-state $k^*$and slower capital deepening. The investment gap, Investment per worker stays relatively close to Depreciation for a longer period compared to China. In the Solow framework, if investment is

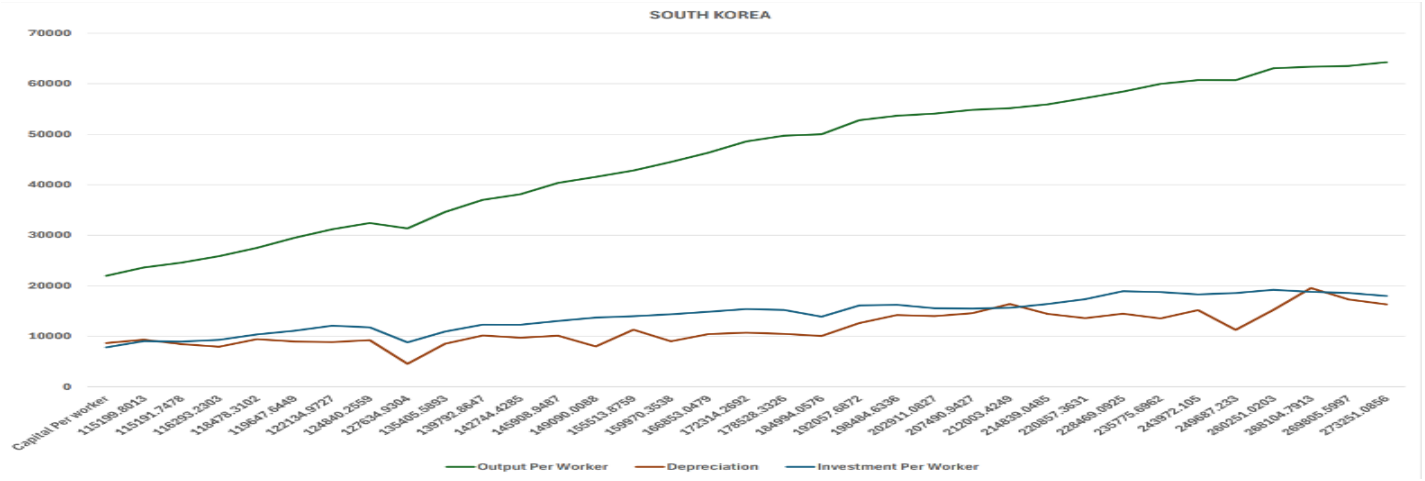
only slightly higher than depreciation, capital per worker grows very slowly, leading to the "slow-growth" trap India experienced for decades. The literature suggests India's growth is often driven by services and productivity (TFP) rather than the massive physical factory-building (Capital) seen in China. Since your graphs focus on **Capital (k)**, India looks "slower" because it isn't following the "heavy industry/high investment" path of its neighbours.

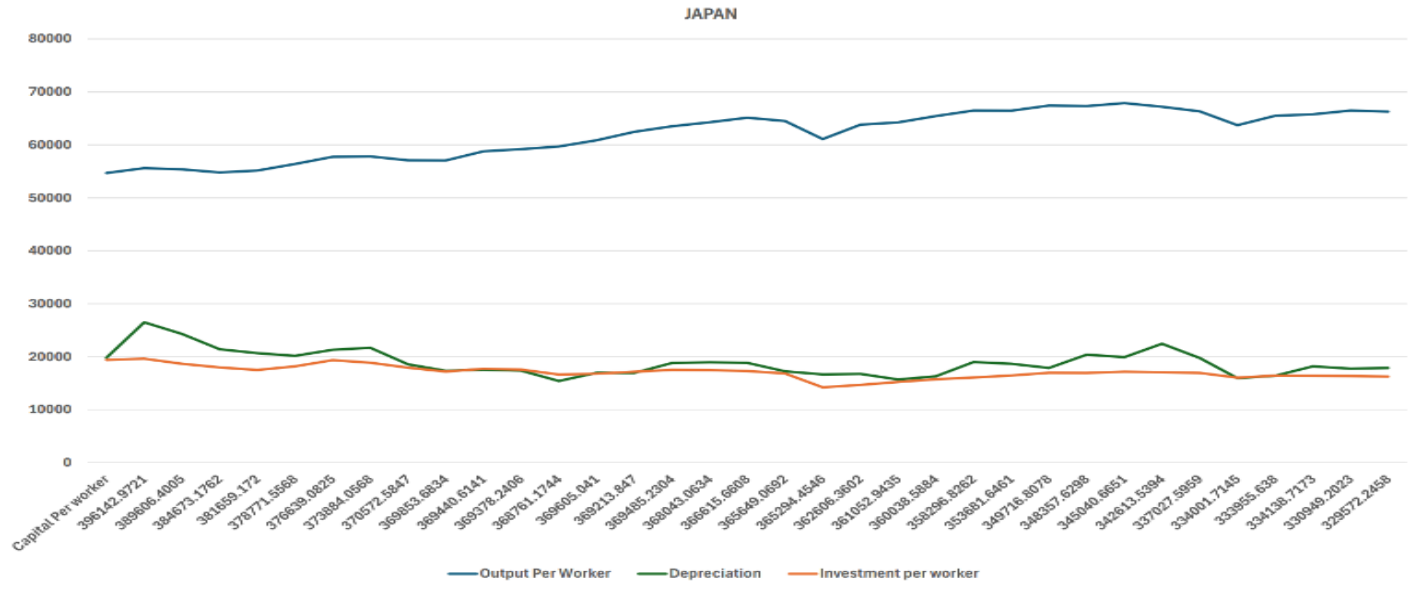
Why Graphs Do Not Follow the Solow Model

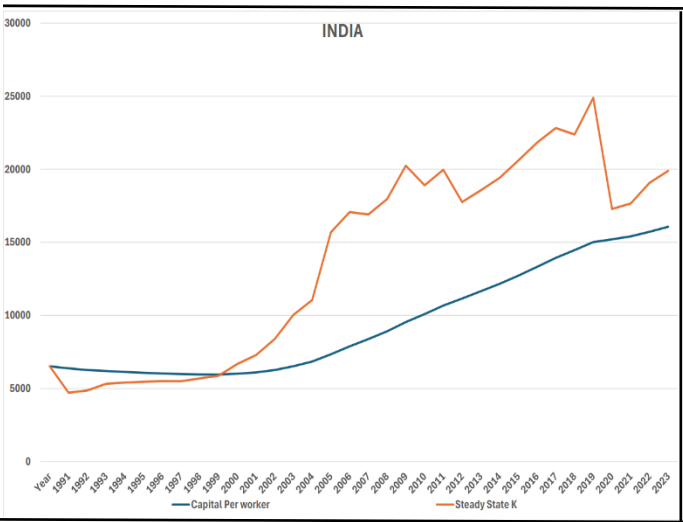
- **Continuous Steady-State Shifts:** In a basic model, the steady state is a fixed point. In your graphs for China and India, the "Steady State k" line is constantly moving upward. This suggests that these countries are not moving toward a static goal, but rather their saving rates (s) or technology levels (A) are improving faster than they can accumulate capital.
- **Volatility and Overshooting:** The Solow model assumes stable growth. However, your graphs for **Japan** and **South Korea** show "Steady State k" lines that fluctuate wildly and sometimes fall below actual capital per worker. This is often due to external shocks (like the 1997 Asian Financial Crisis or the 2008 global crash) which the basic Solow model cannot explain without adding complex variables.
- **Technology is not constant:** Solow assumes exogenous and constant technology but in reality China, Japan, and South Korea experienced rapid technological progress and India adopted technology more slowly.

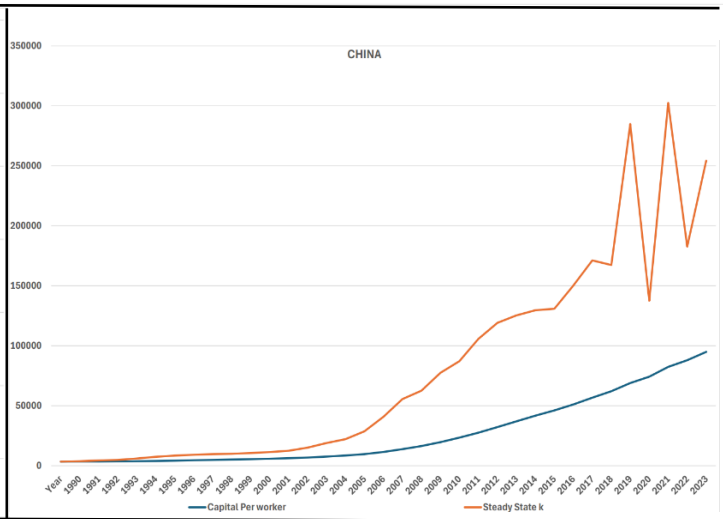


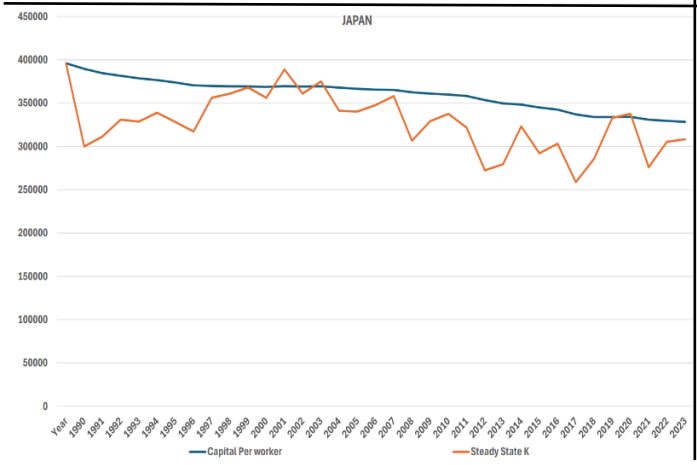


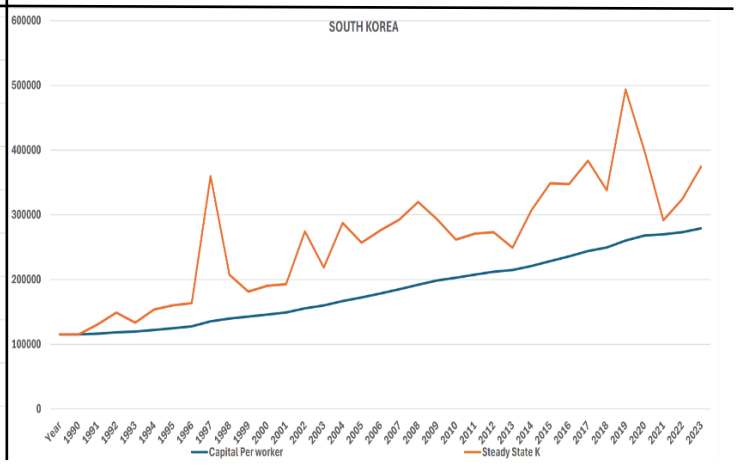

SOUTH KOREA
70000
60000
50000
40000
30000
20000
10000
0
Capital Per worker
115199.8913
115191.7478
118293.2303
118478.3102
119647.6449
122134.9727
124840.2558
127634.9304
135405.5893
139792.8847
142744.4285
145908.9487
149090.0898
155513.8759
159670.3538
166853.0479
172314.2692
178528.3326
184994.0576
192057.6872
198484.6336
202911.6827
207490.8427
212003.4249
214839.6485
220857.3631
228469.0925
235775.6962
243972.105
249667.233
260251.0203
268104.7313
269805.5997
273251.0856
Output Per Worker
Depreciation
Investment Per Worker


JAPAN
80000
70000
60000
50000
40000
30000
20000
10000
0
Capital Per worker
396142.9721
389606.4005
384673.1762
381659.172
378771.5568
376639.0825
373884.0568
370572.5847
369853.6834
369440.6141
369378.2406
368761.1744
369605.041
369213.847
369485.2304
368043.0634
366615.6608
365649.0692
365294.4546
362606.3602
361052.9435
360038.5884
358296.8262
353681.6461
349716.8078
348357.6298
345040.6651
342613.5394
337027.5959
334001.7145
333855.638
334138.7173
330949.2023
329572.2458
Output Per Worker
Depreciation
Investment per worker


INDIA
30000
25000
20000
15000
10000
5000
0
Year
Capital Per worker
Steady State K


CHINA
350000
300000
250000
200000
150000
100000
50000
0
Year
Capital Per worker
Steady State k


JAPAN
450000
400000
350000
300000
250000
200000
150000
100000
50000
0
Year
Capital Per worker
Steady State K


SOUTH KOREA
600000
500000
400000
300000
200000
100000
0
Year
Capital Per worker
Steady State K